Exploring Daily Weather Patterns in State College, Pennsylvania

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1 Introduction

This report explores daily weather patterns in State College, Pennsylvania using daily station-level weather records from NOAA Climate Data Online. The purpose of the analysis is to understand how local weather varies across years, months, and seasons. The project focuses on maximum temperature, minimum temperature, precipitation, snowfall, and snow depth.

The main research question is: **What patterns can we observe in daily weather in State College, Pennsylvania over the last ten years?** More specifically, the analysis examines whether State College shows clear seasonal temperature patterns, which months tend to receive more precipitation, and how snowfall varies across years.

2 Data Provenance

The data used in this report come from NOAA Climate Data Online, specifically the Global Historical Climatology Network Daily data for the State College, Pennsylvania weather station. The station used for this project is **STATE COLLEGE, PA US**, station ID **GHCND:USC00368449**. The working file for this analysis is `SC_weather_10.csv`.

The dataset contains daily observations. Important original variables include daily maximum temperature (`TMAX`), daily minimum temperature (`TMIN`), precipitation (`PRCP`), snowfall (`SNOW`), and snow depth (`SNWD`). Additional variables were created during data cleaning, including `year`, `month`, `season`, `temp_range`, `rain_day`, `snow_day`, `heavy_prcp_day`, and `freezing_day`.

Because the data come from NOAA, a public scientific data provider, the dataset is appropriate for reproducible exploratory data analysis. However, the analysis should still be interpreted carefully because 2016 and 2026 are partial years in the current file, rather than complete calendar years.

3 FAIR and CARE Principles

The dataset generally satisfies the **FAIR principles**. It is findable because it can be located through NOAA Climate Data Online using the station ID. It is accessible because NOAA provides public access to daily weather station data. It is interoperable because the data are

stored in a standard CSV format and use common weather variables. It is reusable because the data include clear variable names, station information, and measurement records that can be analyzed in R.

The **CARE principles** are less directly applicable because this project does not use Indigenous data or community-owned cultural data. However, the project still follows the spirit of responsible data use by clearly describing the source, avoiding unsupported causal claims, and using the data only for exploratory weather analysis. The report also acknowledges NOAA as the original data provider and maintains transparency about created variables and partial-year observations.

4 Exploratory Data Analysis

4.1 Annual Weather Summary

Table 1 summarizes the annual weather records. Complete years generally contain 365 or 366 days, while 2016 and 2026 are partial years in this file. For that reason, comparisons involving total precipitation or total snowfall should focus mainly on complete years.

Table 1: Annual weather summary for State College, PA.

Year	Days observed	Avg. Maximum Temp	Avg. Minimum Temp	Avg. Daily Temp Range	Total Precipitation	Total Snowfall	Max Snow Depth	Rain Days	Snow Days	Freezing Days
2016	256	67.6	50.5	17.1	24.89	7.0	4	102	7	36
2017	365	59.4	43.3	16.2	42.19	37.7	10	161	23	114
2018	365	57.9	42.8	15.1	63.75	39.1	11	184	24	131
2019	365	59.0	42.4	16.6	41.50	32.5	7	155	23	128
2020	366	60.9	43.9	17.0	39.10	25.2	13	151	19	110
2021	365	61.7	44.0	17.7	45.94	32.3	12	157	31	121
2022	365	60.8	41.7	19.1	39.80	43.0	5	147	26	130
2023	365	61.9	44.0	18.0	39.97	11.1	2	154	16	102
2024	366	61.6	44.4	17.2	43.18	28.0	5	144	19	100
2025	365	59.6	42.3	17.3	38.36	21.8	4	142	28	122
2026	110	45.0	27.2	17.9	9.17	20.5	10	40	11	67

The table shows that annual average maximum temperatures for complete years are mostly near the upper 50s to low 60s in degrees Fahrenheit. The largest annual precipitation total appears in 2018, while 2022 has the largest total snowfall among the complete years shown.

4.2 Annual Temperature Pattern

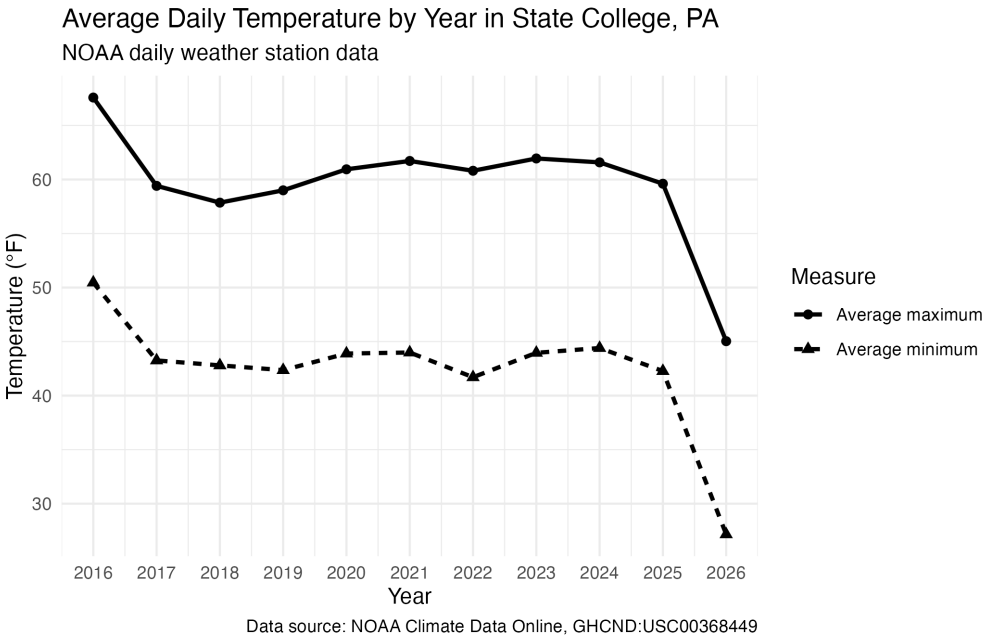

Figure 1: Line graph showing average annual maximum and minimum temperatures in State College, Pennsylvania. The two lines show that complete-year average maximum temperatures usually stay near the upper 50s to low 60s, while average minimum temperatures usually stay in the low to mid 40s. The partial year 2026 appears much colder because it does not include a full calendar year.

Figure 1 shows that the annual temperature pattern is relatively stable for complete years. Average maximum temperature usually remains close to 60°F, while average minimum temperature is generally in the low to mid 40s. The sharp drop in 2026 should not be interpreted as a long-term cooling pattern because 2026 is only a partial year in the dataset.

4.3 Monthly Weather Summary

Table 2 compares monthly averages and totals across the full dataset. This table helps show the seasonal structure of weather in State College.

Table 2: Monthly weather summary for State College, PA.

Month	Avg. Maximum Temp	Avg. Minimum Temp	Median Daily Precipitation	Total Precipitation	Total Snowfall	Rain Days	Snow Days
January	35.4	22.1	0.00	28.07	88.9	137	65
February	40.5	24.0	0.00	29.08	76.4	113	58
March	49.0	30.3	0.00	30.71	44.3	134	33
April	60.5	41.1	0.01	36.72	9.6	159	6
May	69.1	51.2	0.01	48.86	0.0	160	0
June	77.7	59.7	0.00	38.98	0.0	125	0
July	83.3	65.3	0.00	42.65	0.0	127	0
August	80.6	63.0	0.00	43.21	0.0	124	0
September	74.3	55.9	0.00	37.83	0.0	108	0
October	63.6	45.1	0.00	34.70	0.0	119	0
November	49.8	33.2	0.00	26.01	21.9	97	18
December	39.5	26.5	0.00	31.03	57.1	134	47

The monthly table shows a clear seasonal temperature cycle. July has the highest average maximum temperature, while January has the lowest average maximum and minimum temperatures. Snowfall is concentrated in winter and early spring months, especially January, February, March, and December.

4.4 Monthly Precipitation

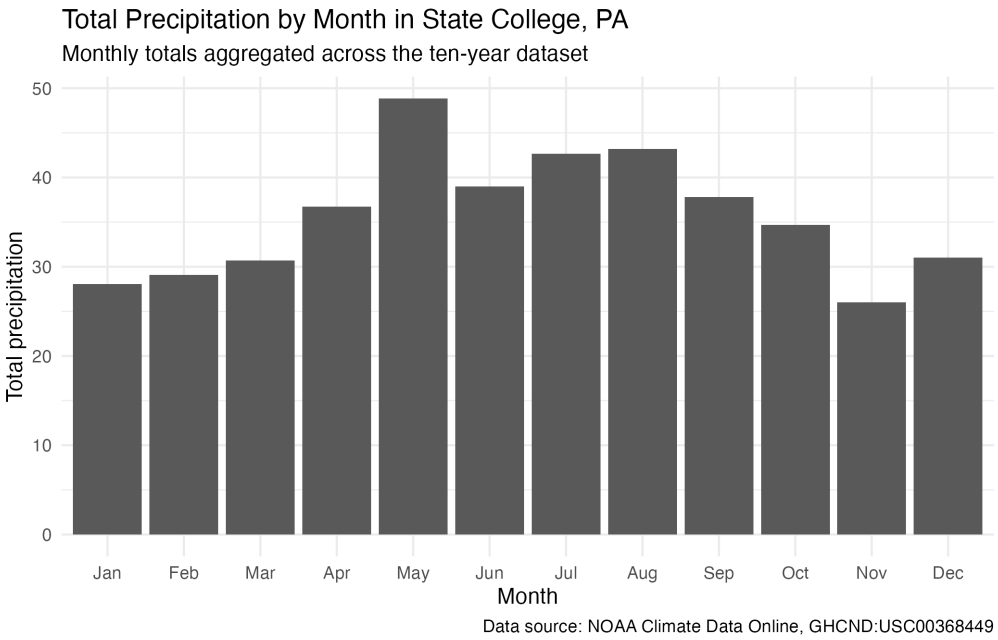

Figure 2: Bar chart showing total precipitation by month in State College, Pennsylvania. May has the highest total precipitation, while November has one of the lowest totals. Precipitation occurs in all months, but the total amount varies by season.

Figure 2 shows that precipitation is not limited to one season. May has the highest total precipitation in the summarized dataset, while late fall and winter months tend to be lower than spring and summer months. This supports the idea that rainfall and precipitation in State College occur throughout the year but vary by month.

4.5 Distribution of Daily Precipitation

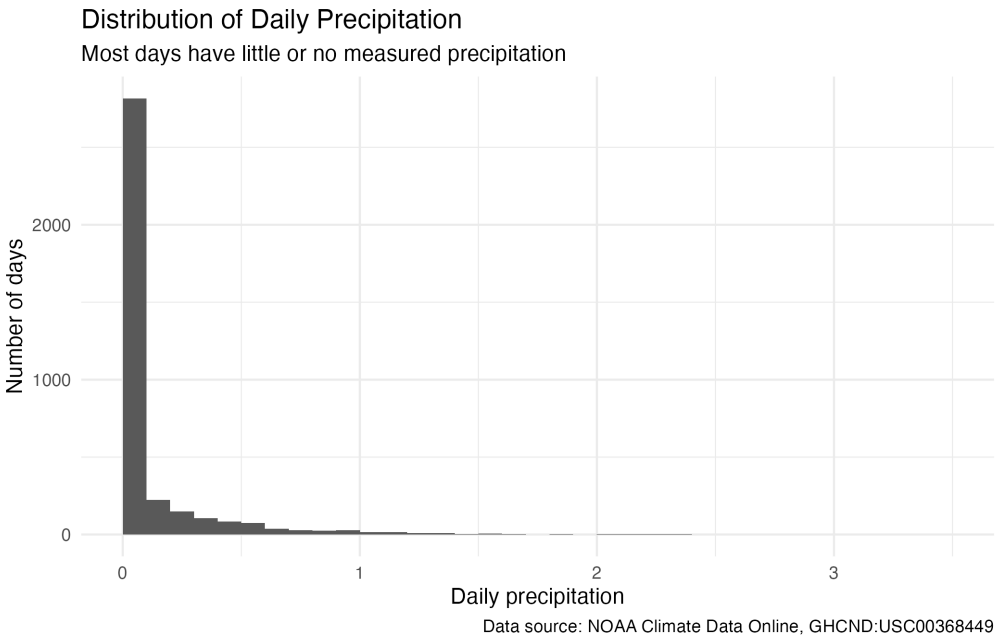

Figure 3: Histogram showing the distribution of daily precipitation. Most observations are near zero, with a long right tail for days with heavier precipitation.

Figure 3 shows that daily precipitation is strongly right-skewed. Most days have little or no precipitation, while a small number of days have much larger precipitation values. This explains why the median daily precipitation in many months is 0.00 even though the monthly total precipitation is not zero.

4.6 Seasonal Temperature Distribution

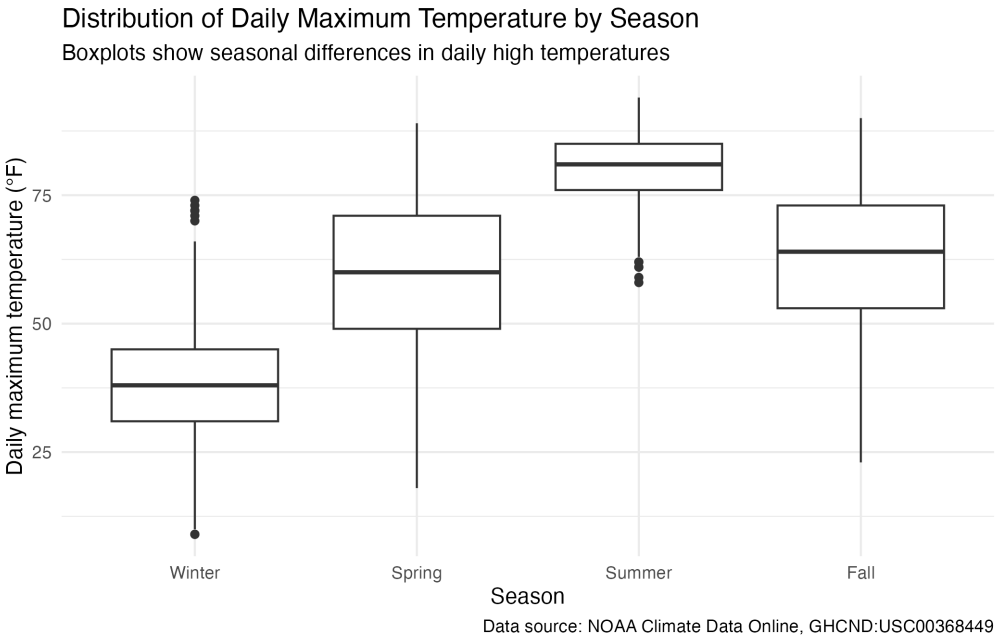

Figure 4: Boxplots showing the distribution of daily maximum temperature by season. Summer has the highest median and the narrowest warm-season range, while winter has the lowest median and several warmer outliers. Spring and fall show wider transition-season variation.

Figure 4 shows that daily maximum temperatures differ strongly by season. Summer has the highest median daily maximum temperature, while winter has the lowest. Spring and fall show broader variation because they are transition seasons.

4.7 Annual Snowfall

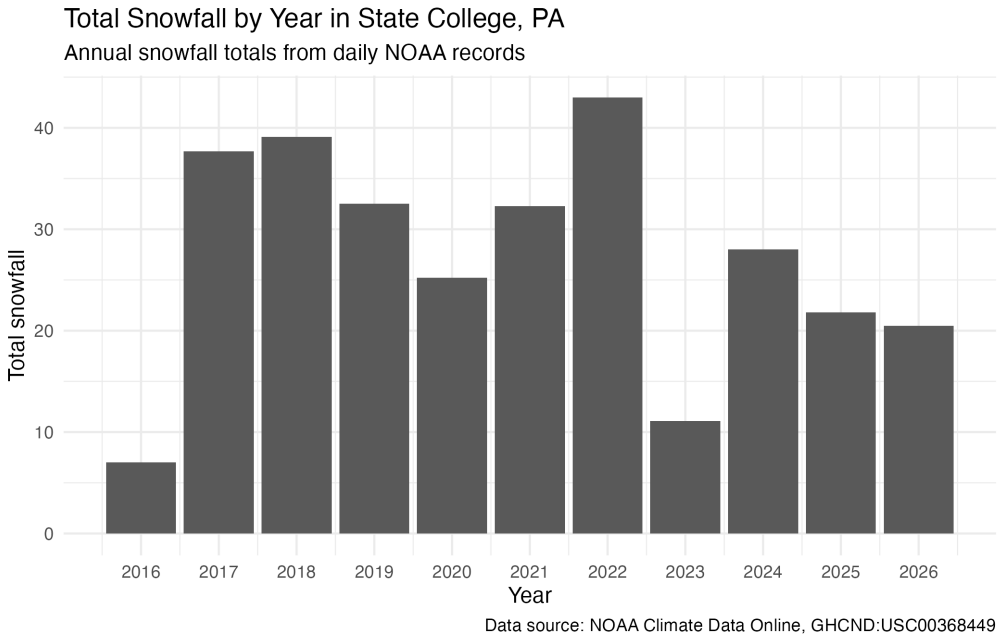

Figure 5: Bar chart showing total snowfall by year in State College, Pennsylvania. Snowfall varies substantially by year, with 2022 among the highest complete years and 2023 among the lowest complete years.

Figure 5 shows that snowfall varies substantially from year to year. Among the complete years, 2022 has the highest total snowfall, while 2023 has a much lower snowfall total. This suggests that winter weather in State College can vary considerably even when annual average temperatures are relatively stable.

4.8 Monthly Temperature Distribution

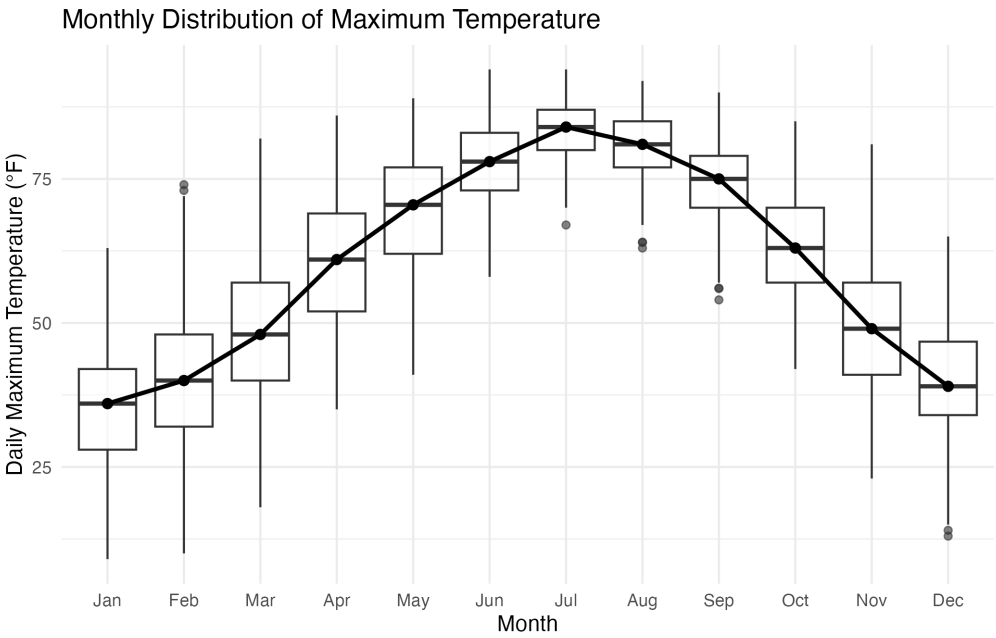

Figure 6: Boxplots showing the monthly distribution of daily maximum temperature in State College, PA. The line connects monthly median values and shows the overall seasonal temperature trend. Daily maximum temperatures rise from winter to summer, peak in July, and decline through fall.

Figure 6 shows that daily maximum temperatures follow a strong seasonal pattern. Temperatures are lowest in January and February, rise through spring, peak in July, and then decline through fall and early winter. The boxplots also show that temperature variation differs by month. Transitional months such as March, April, October, and November show wider ranges because they include both colder and warmer days. The median trend line makes the annual temperature cycle easier to see while the boxplots preserve the daily-level variation within each month.

4.9 Rainfall and Temperature Diagram

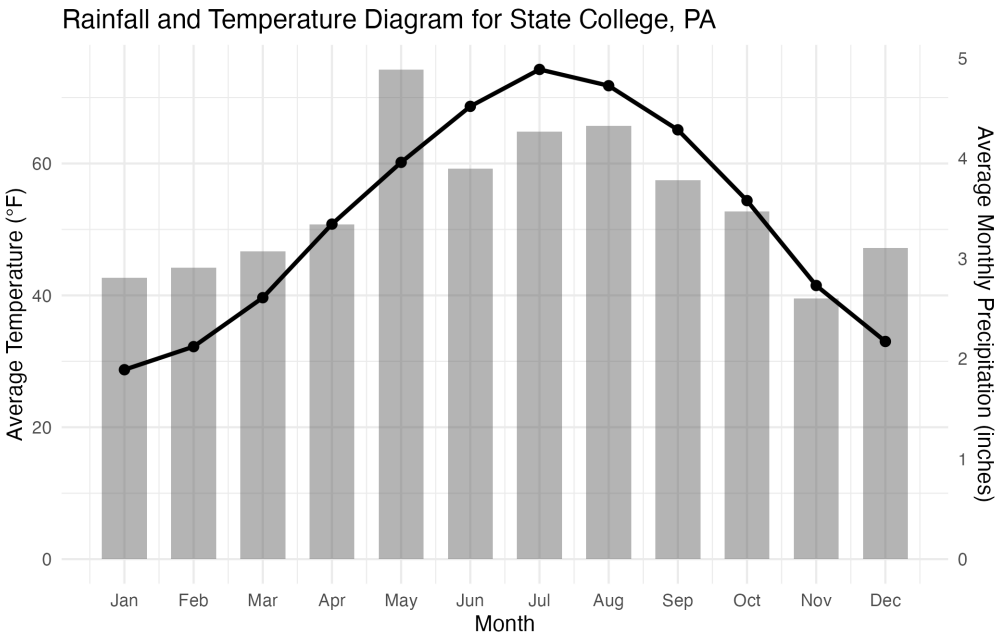

Figure 7: Rainfall and temperature diagram for State College, PA. The bars show average monthly precipitation, while the line shows average monthly temperature.

Figure 7 shows that State College has both a clear annual temperature cycle and year-round precipitation. Average temperature increases from winter to summer, reaches its highest level in July, and then decreases during fall and winter. Precipitation occurs in every month, with relatively higher values in late spring and summer. This pattern suggests rain-heat synchronization because warmer months also receive substantial precipitation. Overall, the diagram supports the interpretation that State College has a humid climate pattern with warm to hot summers, cold winters, and no obvious dry season.

4.10 Time Series Analysis

4.10.1 Monthly Within-Cycle Variation Summary

Table 3: Monthly summary of within-cycle variation in temperature, precipitation, rain frequency, and freezing conditions.

Month	Mean TMAX	SD TMAX	Mean TMIN	SD TMIN	Mean PRCP	Rain Day Rate	Freezing Day Rate
Jan	35.4	10.7	22.1	11.2	0.091	44.2%	83.2%
Feb	40.5	11.8	24.0	9.23	0.103	40.1%	83.7%
Mar	49.0	12.1	30.3	8.90	0.099	43.4%	61.3%
Apr	60.5	12.2	41.1	8.74	0.122	52.8%	20.6%
May	69.1	10.4	51.2	8.06	0.158	51.6%	0.6%
Jun	77.6	7.15	59.7	6.41	0.130	41.7%	0.0%
Jul	83.3	4.91	65.3	4.74	0.138	41.0%	0.0%
Aug	80.6	5.80	63.0	5.39	0.139	40.0%	0.0%
Sep	74.3	7.24	55.9	7.01	0.126	36.0%	0.0%
Oct	63.6	9.50	45.1	8.60	0.112	38.4%	3.9%
Nov	49.8	11.0	33.2	7.42	0.087	32.3%	52.3%
Dec	39.5	9.50	26.5	8.29	0.100	43.2%	78.7%

The monthly summary shows a strong seasonal cycle in temperature. Maximum temperature increases steadily from winter to summer, peaking in July at 83.3°F, before declining again toward winter. Summer also has the smallest standard deviation, while spring and fall show wider variation. Precipitation remains present throughout the year, with the highest mean daily precipitation in May. Rain day frequency remains relatively high across all months, supporting the interpretation of a humid climate without a clear dry season.

4.10.2 Within-Cycle Variation in Maximum Temperature

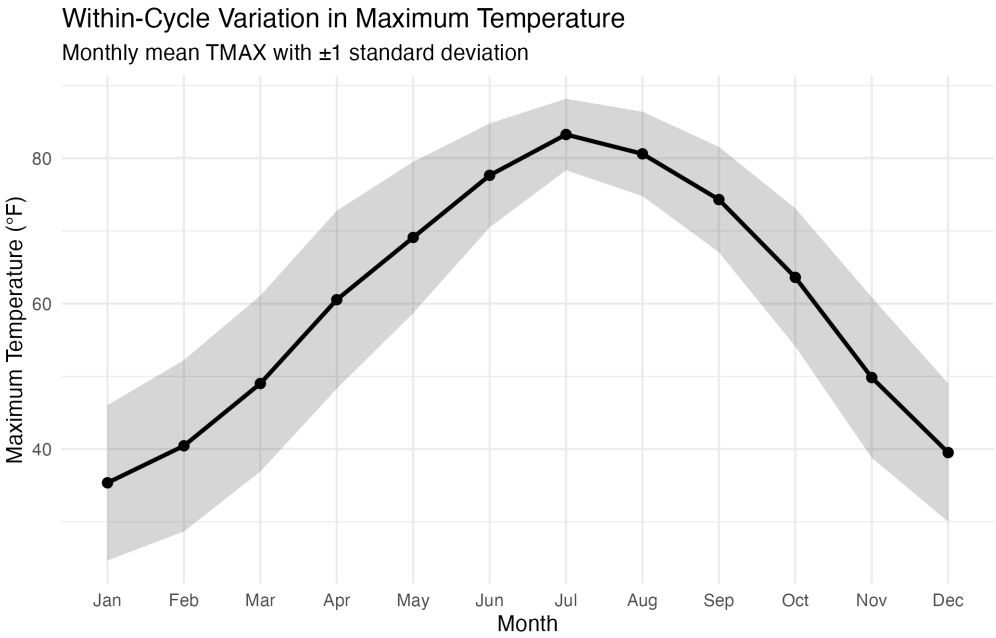

Figure 8: Line chart showing monthly mean daily maximum temperature with a ±1 standard deviation band. Temperatures rise from winter to summer, peak in July, and decline through fall. The variation band is wider in spring and fall, showing stronger transition-season variability.

Figure 8 shows a clear annual temperature cycle. Mean maximum temperature rises from about 35°F in January to about 83°F in July, then declines again toward winter. The shaded standard deviation band shows that spring and fall have broader variation, while summer is relatively more stable. This indicates strong within-cycle variation driven by seasonal temperature change.

4.10.3 Monthly Precipitation and Rain Day Frequency

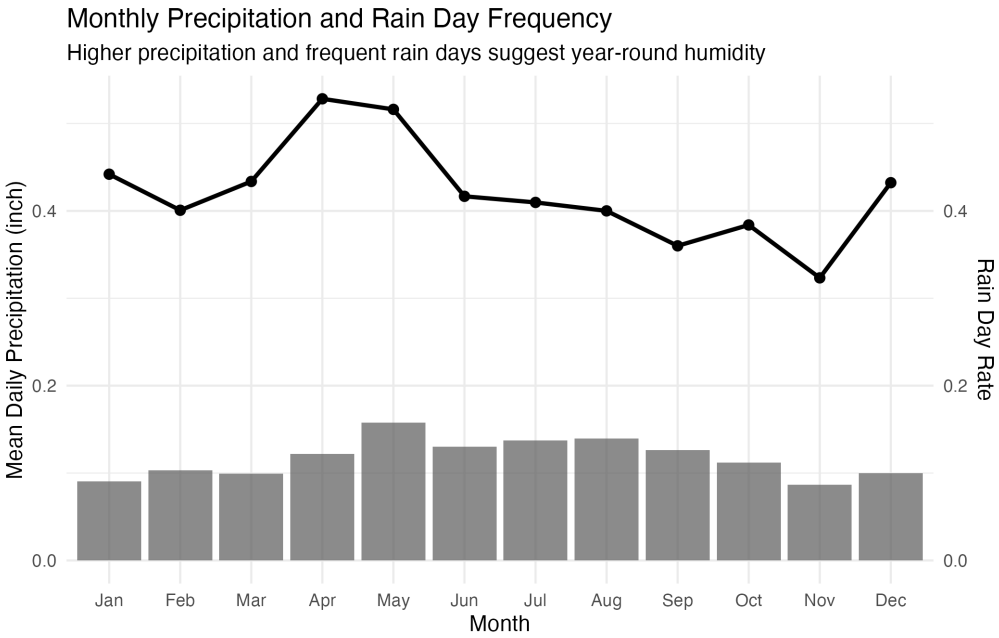

Figure 9: Bar and line chart showing monthly mean daily precipitation and rain day frequency. Precipitation occurs throughout the year, with slightly higher values in late spring and summer. Rain day frequency remains substantial across all months.

Figure 9 shows that precipitation is distributed across the full year rather than concentrated in a single wet season. Mean daily precipitation is highest around May and remains relatively high through summer. Rain day frequency also stays substantial in every month, which supports the interpretation of a humid climate with no clear dry season. The overlap between warmer months and higher precipitation also suggests rain-heat synchrony.

4.10.4 Annual Between-Cycle Variation Summary

Table 4: Annual summary of between-cycle variation across complete years only.

Year	Mean TMAX	Mean TMIN	SD TMAX	Total PRCP	Total Snow	Rain Days	Snow Days	Freezing Days
2017	59.4	43.3	18.7	42.2	37.7	161	23	114
2018	57.9	42.8	20.3	63.8	39.1	184	24	131
2019	59.0	42.4	19.4	41.5	32.5	155	23	128

Year	Mean TMAX	Mean TMIN	SD TMAX	Total PRCP	Total Snow	Rain Days	Snow Days	Freezing Days
2020	60.9	43.9	18.4	39.1	25.2	151	19	110
2021	61.7	44.0	18.7	45.9	32.3	157	31	121
2022	60.8	41.7	19.4	39.8	43.0	147	26	130
2023	61.9	44.0	16.2	40.0	11.1	154	16	102
2024	61.6	44.4	18.0	43.2	28.0	144	19	100
2025	59.6	42.3	19.8	38.4	21.8	142	28	122

The annual summary shows that mean temperature remains relatively stable across years, with most annual mean maximum temperatures staying between 58°F and 62°F. In contrast, total precipitation and snowfall vary much more substantially. For example, 2018 is the wettest year, while 2023 has much lower snowfall. This suggests that temperature is more stable over time, while precipitation and snowfall are more sensitive to year-to-year weather variability.

4.10.5 Between-Cycle Variation in Annual Mean Maximum Temperature

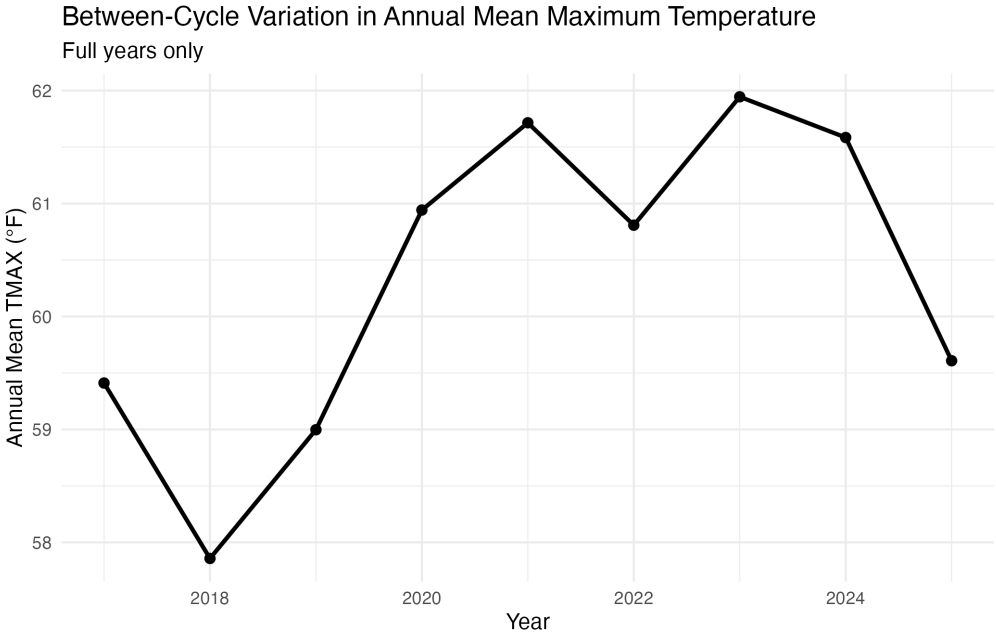

Figure 10: Line chart showing annual mean maximum temperature across complete years. Annual temperature varies moderately, with most years falling between about 58°F and 62°F.

Figure 10 shows that annual mean maximum temperature varies only moderately across complete years. Most values remain within a narrow range of roughly 58°F to 62°F. Compared with the much larger monthly temperature range, this suggests that between-cycle temperature variation is weaker than within-cycle seasonal variation. In other words, the annual cycle is much stronger than year-to-year temperature fluctuation.

4.10.6 Between-Cycle Variation in Annual Precipitation

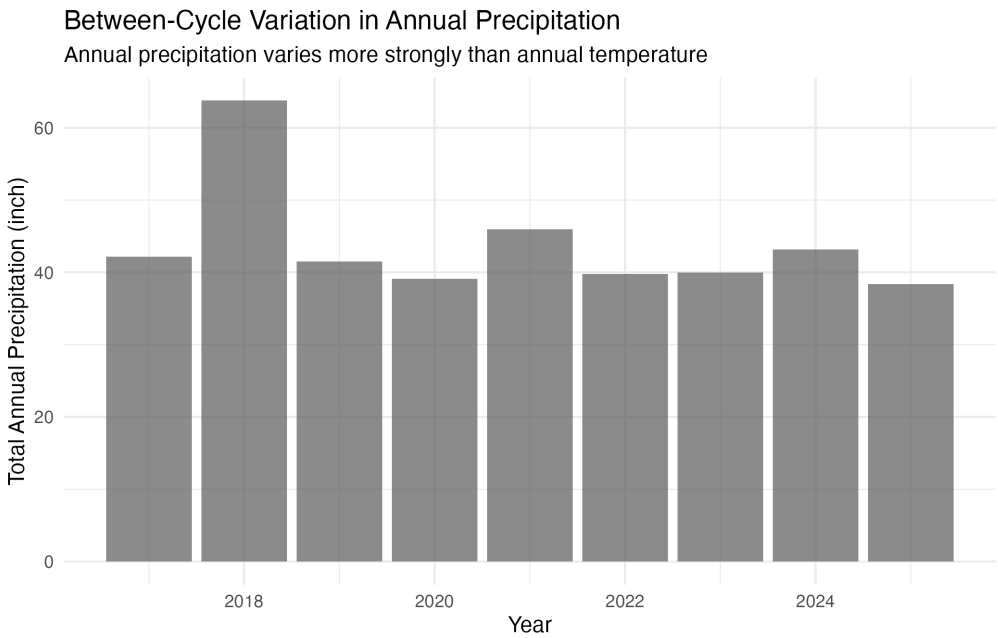

Figure 11: Bar chart showing total annual precipitation across complete years. Annual precipitation varies more strongly than annual mean temperature, with 2018 standing out as the wettest year.

Figure 11 shows that annual precipitation has much stronger between-cycle variation than annual mean temperature. The wettest year is 2018, with total precipitation above 60 inches, while several other complete years are closer to 40 inches. This suggests that precipitation is more irregular across years than temperature, which is expected because rainfall is more strongly affected by storm systems and short-term atmospheric conditions.

4.10.7 Daily Time Series of Maximum Temperature

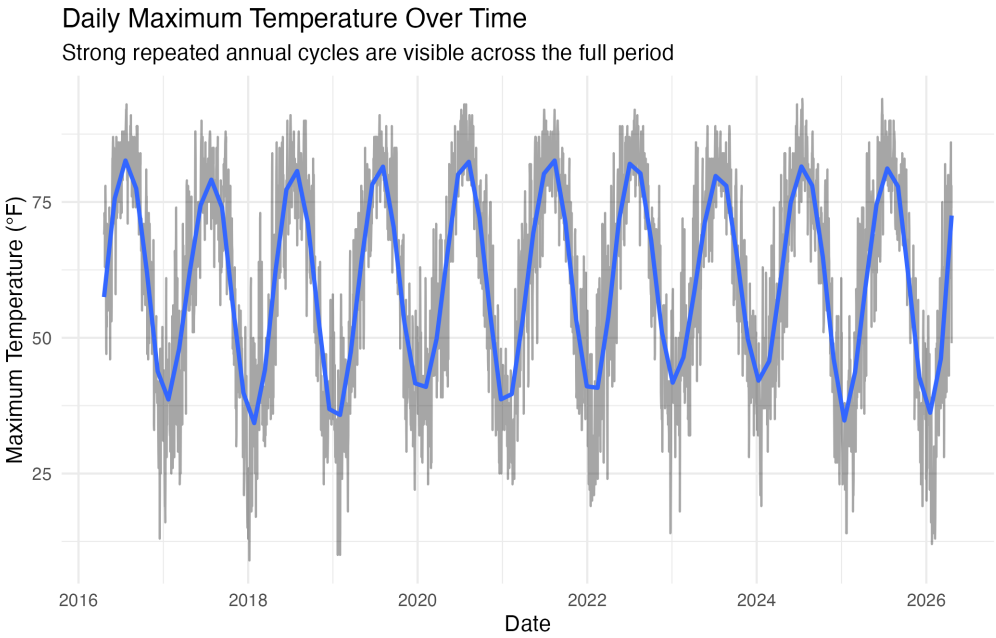

Figure 12: Daily time series plot showing maximum temperature over time. The gray line shows daily variation, while the smoothed blue line shows repeated annual cycles across the full period.

Figure 12 shows both short-term daily fluctuation and repeated annual cycles. The gray line captures day-to-day weather variability, while the smoothed line shows the broader seasonal pattern. The repeated peaks and troughs across years indicate strong seasonality, with high summer temperatures and low winter temperatures recurring consistently over time. This confirms that the main structure of the temperature series is annual periodicity.

4.10.8 Summary Table for Within-Cycle and Between-Cycle Variation

Table 5: Summary comparison of within-cycle and between-cycle variation.

Dimension	Measure	Value
Within-cycle temperature variation	Monthly mean TMAX range	47.9
Within-cycle precipitation variation	Monthly mean PRCP range	0.0709
Between-cycle temperature variation	Annual mean TMAX range	4.09
Between-cycle precipitation variation	Annual total PRCP range	25.4

The summary table confirms that temperature is dominated by strong within-cycle seasonal variation, while precipitation shows stronger between-cycle irregularity. Monthly mean maximum temperature changes by nearly 48°F within a year, but annual mean maximum temperature changes by only about 4°F across years. In contrast, annual total precipitation varies substantially between years. This indicates that temperature follows a stable annual cycle, while precipitation is more sensitive to interannual variation.

5 Overall Data Story

The exploratory analysis shows that State College has a strong seasonal weather pattern and a typical humid continental climate structure. Temperatures are lowest in winter, rise through spring, peak in summer, and decline through fall. Both the seasonal boxplots and the monthly temperature distribution confirm that summer has the highest daily maximum temperatures, while winter has the lowest. Spring and fall show broader variation because they are transition seasons between cold and warm conditions.

The time series analysis further confirms that temperature is dominated by strong within-cycle seasonal variation. Monthly mean maximum temperature increases from about 35°F in January to more than 83°F in July before declining again toward winter, creating a clear annual cycle. The standard deviation bands show that summer temperatures are relatively stable, while spring and fall have wider variation due to rapid seasonal transitions. Daily temperature series across the full period also show repeated annual peaks and troughs, indicating that annual periodicity is the main structure of the temperature pattern.

Precipitation occurs throughout the year rather than being concentrated in a single season. The monthly precipitation totals and the rainfall–temperature diagram show that late spring and summer months contribute more precipitation, while winter precipitation remains lower but is often associated with snowfall. This indicates a pattern of rain–heat synchronization, where warmer months tend to receive relatively more rainfall. Rain day frequency also remains relatively high across all months, which supports the classification of a fully humid climate without a clear dry season.

Snowfall is concentrated mainly in winter and early spring, especially from December through March, but the total annual snowfall varies substantially across years. Some years, such as 2022, show much heavier snowfall than others, while annual average temperatures remain relatively stable. This suggests that winter severity can change considerably even without large shifts in annual mean temperature.

The time series comparison between within-cycle and between-cycle variation shows that seasonal temperature change is much stronger than year-to-year temperature fluctuation. Monthly mean maximum temperature varies by nearly 48°F within a year, while annual mean maximum temperature across complete years varies by only about 4°F. In contrast, annual

total precipitation varies much more strongly across years, with 2018 standing out as an especially wet year. This suggests that temperature follows a stable long-term seasonal cycle, while precipitation is more sensitive to interannual atmospheric variability.

The analysis also highlights the importance of distinguishing daily distributions from monthly or annual totals. For example, daily precipitation is strongly right-skewed, meaning that most days have little or no precipitation, while a smaller number of heavy precipitation events contribute substantially to monthly totals. Similarly, monthly boxplots show that average seasonal trends can hide substantial day-to-day variation within each month.

The partial-year observations for 2016 and 2026 are important context. These years should not be compared directly with complete years when interpreting total precipitation, total snowfall, or annual average temperature.

6 Open Science and Reproducibility

This project follows open science principles by using a public data source, documenting the data provenance, keeping analysis code in a reproducible R workflow, and reporting created variables clearly. The final project repository should include the raw CSV file, the QMD report, generated figures, citation files, and any scripts used to clean and analyze the data.

The project also applies PCIP by making the report readable for a broad audience, using clear captions and alt text, and avoiding unsupported claims. The analysis is exploratory rather than causal, so the report focuses on describing patterns rather than claiming that one weather pattern causes another.

7 Author Contributions

- **Ata:** Contributed to project planning, data interpretation, and review of weather summaries.
- **Jincheng:** Contributed to table review, figure interpretation, time-series analysis and report organization.
- **Qihaohan:** Primary author for the R data cleaning and analysis workflow; contributed to figure generation and report writing.

8 Code Appendix

The following code was used to read, clean, summarize, and visualize the data. Code is placed in the appendix so that the main body of the report focuses on interpretation.

```
# A tibble: 12 x 10
   month_num month mean_tmax sd_tmax mean_tmin sd_tmin mean_prcp sd_prcp
       <dbl> <fct>     <dbl>   <dbl>     <dbl>   <dbl>     <dbl>   <dbl>
 1         1 Jan        35.4   10.7       22.1   11.2     0.0905   0.224
 2         2 Feb        40.5   11.8       24.0    9.23    0.103    0.236
 3         3 Mar        49.0   12.1       30.3    8.90    0.0994   0.236
 4         4 Apr        60.5   12.2       41.1    8.74    0.122    0.259
 5         5 May        69.1   10.4       51.2    8.06    0.158    0.294
 6         6 Jun        77.6    7.15      59.7    6.41    0.130    0.292
 7         7 Jul        83.3    4.91      65.3    4.74    0.138    0.305
 8         8 Aug        80.6    5.80      63.0    5.39    0.139    0.355
 9         9 Sep        74.3    7.24      55.9    7.01    0.126    0.353
10        10 Oct        63.6    9.50      45.1    8.60    0.112    0.272
11        11 Nov        49.8   11.0       33.2    7.42    0.0867   0.271
12        12 Dec        39.5    9.50      26.5    8.29    0.100    0.261
# i 2 more variables: rain_day_rate <dbl>, freezing_day_rate <dbl>
```

```
# A tibble: 9 x 10
   year mean_tmax mean_tmin sd_tmax total_prcp total_snow rain_days snow_days
  <dbl>     <dbl>     <dbl>   <dbl>      <dbl>      <dbl>     <int>     <int>
1  2017      59.4      43.3    18.7       42.2       37.7       161        23
2  2018      57.9      42.8    20.3       63.8       39.1       184        24
3  2019      59.0      42.4    19.4       41.5       32.5       155        23
4  2020      60.9      43.9    18.4       39.1       25.2       151        19
5  2021      61.7      44.0    18.7       45.9       32.3       157        31
6  2022      60.8      41.7    19.4       39.8       43         147        26
7  2023      61.9      44.0    16.2       40.0       11.1       154        16
8  2024      61.6      44.4    18.0       43.2       28         144        19
9  2025      59.6      42.3    19.8       38.4       21.8       142        28
# i 2 more variables: freezing_days <int>, n_days <int>
```

```
# A tibble: 4 x 3
  dimension                              measure                    value
  <chr>                                  <chr>                      <dbl>
1 Within-cycle temperature variation     Monthly mean TMAX range 47.9
2 Within-cycle precipitation variation   Monthly mean PRCP range  0.0709
3 Between-cycle temperature variation    Annual mean TMAX range   4.09
4 Between-cycle precipitation variation  Annual total PRCP range 25.4
```